

## VECTORS

### 1.1 Learning outcomes

You should be able to use the following concepts to solve problems:

- Position vectors
- Vector addition
- Magnitude (modulus) of a vector
- Dot (scalar) product
- Projection of a vector
- Components of a vector.

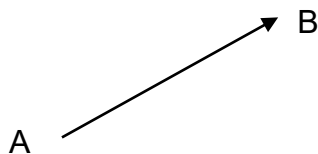
You should be able to calculate the direction of a vector.

If you have previously learnt about matrices, you should be able to represent a vector by a matrix and use matrix multiplication to calculate the dot product.

If you are good at matrix algebra you should learn how to use a matrix to rotate a vector.

### 1.2 Position vector

A *position vector* is a fixed vector from a point A to a point B that defines the position of A relative to B.



It is conventional to represent a position vector in the form  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ .

Here  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  are vectors of unit length that define an orthogonal Cartesian coordinate system (i.e.  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  are mutually perpendicular). Then  $x$ ,  $y$  and  $z$  are called the *components* of  $\mathbf{r}$ . Note that  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  define the  $x$ ,  $y$  and  $z$ -axes of the coordinate system.

### 1.3 Vector addition

To add two vectors we simply add their components, since  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  are orthogonal.

For example, let  $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$  and  $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k}$ ,

then their sum is given by  $\mathbf{c} = \mathbf{a} + \mathbf{b} = (a_x + b_x)\mathbf{i} + (a_y + b_y)\mathbf{j} + (a_z + b_z)\mathbf{k}$ .

### 1.4 Magnitude (modulus) of a vector

Consider the vector  $\mathbf{r}$  defined by  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ .

By Pythagoras' theorem, its *magnitude* is given by  $r = \sqrt{x^2 + y^2 + z^2}$ .

The magnitude of a vector is also called its *modulus*.

### 1.5 Dot (scalar) product

The *dot product* of two vectors,  $\mathbf{a}$  and  $\mathbf{b}$ , is defined by  $\mathbf{a} \cdot \mathbf{b} = ab \cos \vartheta$ ,

where  $\vartheta$  is the angle between  $\mathbf{a}$  and  $\mathbf{b}$ . The dot product is also called the *scalar product* because it produces a scalar result.

From this definition, it follows that the dot product is commutative i.e.  $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ .

It also follows that  $\mathbf{i} \cdot \mathbf{i} = 1$   $\mathbf{j} \cdot \mathbf{j} = 1$   $\mathbf{k} \cdot \mathbf{k} = 1$ .

Similarly all products between different orthogonal unit vectors are zero; for example  $\mathbf{i} \cdot \mathbf{j} = 0$   $\mathbf{j} \cdot \mathbf{k} = 0$   $\mathbf{k} \cdot \mathbf{i} = 0$  etc.

If  $\mathbf{a}$  and  $\mathbf{b}$  are defined by  $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$  and  $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k}$ ,

it follows, from the above, that  $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$ .

The dot product is used to:

- find the angle between two vectors
- determine the components of a vector
- define the work done by a force.

It follows, from the definition of the dot product, that the angle between two vectors  $\mathbf{a}$  and  $\mathbf{b}$  is given by

$$\vartheta = \cos^{-1} \left( \frac{\mathbf{a} \cdot \mathbf{b}}{ab} \right).$$

### 1.6 Projection of a vector

The *projection* of a vector  $\mathbf{a}$  in the direction defined by a vector  $\mathbf{b}$  (i.e. the projection of  $\mathbf{a}$  on the direction given by  $\mathbf{b}$ ) is given by

$$c = a \cos \vartheta = \frac{ab \cos \vartheta}{b} \text{ where } \vartheta \text{ is the angle between } \mathbf{a} \text{ and } \mathbf{b}.$$

### 1.7 Components of a vector

The vector  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  can be written in the form

$$\mathbf{r} = (r \cos \alpha)\mathbf{i} + (r \cos \beta)\mathbf{j} + (r \cos \gamma)\mathbf{k}.$$

where  $\mathbf{r}$  makes angles of  $\alpha$ ,  $\beta$  and  $\gamma$  with  $\mathbf{i}$ ,  $\mathbf{j}$  and  $\mathbf{k}$  respectively. Then the angles  $\alpha$ ,  $\beta$  and  $\gamma$  define the direction of  $\mathbf{r}$ ; their cosines are called the *direction cosines* of  $\mathbf{r}$ .

Since  $\mathbf{i}$  has unit length,  $r \cos \alpha = \mathbf{r} \cdot \mathbf{i} \Rightarrow \alpha$ . By determining  $\alpha$ ,  $\beta$  and  $\gamma$ , in this manner, the direction of  $\mathbf{r}$  can be specified with respect to the coordinate system that they define.

### 1.8 Matrix representation

Consider two vectors  $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$  and  $\mathbf{b} = b_x\mathbf{i} + b_y\mathbf{j} + b_z\mathbf{k}$ .

In matrix notation, these vectors are written as

$$\mathbf{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$

The usual rules of matrix addition give the result

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = \begin{bmatrix} a_x + b_x \\ a_y + b_y \\ a_z + b_z \end{bmatrix}$$

which is equivalent to the result given using vector notation.

The transpose of  $\mathbf{a}$  is written  $\mathbf{a}^T = [a_x \ a_y \ a_z]$ .

Then

$$\mathbf{a}^T \mathbf{b} = [a_x \ a_y \ a_z] \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = a_x b_x + a_y b_y + a_z b_z.$$

Thus, in matrix notation,  $\mathbf{a}^T \mathbf{b}$  gives the dot product of  $\mathbf{a}$  and  $\mathbf{b}$ .

## 10.9 Introductory Examples

### Example 1

#### EXAMPLE 2.5

The link in Fig. 2-18a is subjected to two forces  $F_1$  and  $F_2$ . Determine the magnitude and direction of the resultant force.

##### SOLUTION I

**Scalar Notation.** First we resolve each force into its  $x$  and  $y$  components, Fig. 2-18b, then we sum these components algebraically.

$$\begin{aligned} \rightarrow (F_R)_x = \Sigma F_x; \quad (F_R)_x &= 600 \cos 30^\circ \text{ N} - 400 \sin 45^\circ \text{ N} \\ &= 236.8 \text{ N} \rightarrow \\ + \uparrow (F_R)_y = \Sigma F_y; \quad (F_R)_y &= 600 \sin 30^\circ \text{ N} + 400 \cos 45^\circ \text{ N} \\ &= 582.8 \text{ N} \uparrow \end{aligned}$$

The resultant force, shown in Fig. 2-18c, has a magnitude of

$$\begin{aligned} F_R &= \sqrt{(236.8 \text{ N})^2 + (582.8 \text{ N})^2} \\ &= 629 \text{ N} \end{aligned}$$

*Ans.*

From the vector addition,

$$\theta = \tan^{-1}\left(\frac{582.8 \text{ N}}{236.8 \text{ N}}\right) = 67.9^\circ$$

*Ans.*

##### SOLUTION II

**Cartesian Vector Notation.** From Fig. 2-18b, each force is first expressed as a Cartesian vector.

$$\begin{aligned} \mathbf{F}_1 &= \{600 \cos 30^\circ \mathbf{i} + 600 \sin 30^\circ \mathbf{j}\} \text{ N} \\ \mathbf{F}_2 &= \{-400 \sin 45^\circ \mathbf{i} + 400 \cos 45^\circ \mathbf{j}\} \text{ N} \end{aligned}$$

Then,

$$\begin{aligned} \mathbf{F}_R &= \mathbf{F}_1 + \mathbf{F}_2 = (600 \cos 30^\circ \text{ N} - 400 \sin 45^\circ \text{ N})\mathbf{i} \\ &\quad + (600 \sin 30^\circ \text{ N} + 400 \cos 45^\circ \text{ N})\mathbf{j} \\ &= \{236.8\mathbf{i} + 582.8\mathbf{j}\} \text{ N} \end{aligned}$$

The magnitude and direction of  $\mathbf{F}_R$  are determined in the same manner as before.

**NOTE:** Comparing the two methods of solution, notice that the use of scalar notation is more efficient since the components can be found *directly*, without first having to express each force as a Cartesian vector before adding the components. Later, however, we will show that Cartesian vector analysis is very beneficial for solving three-dimensional problems.

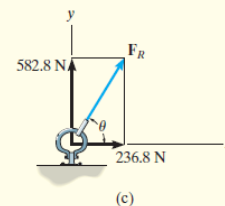
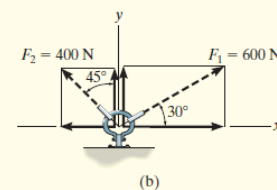
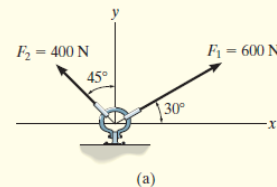


Fig. 2-18

## Example 2

### EXAMPLE 2.8

Express the force  $\mathbf{F}$  shown in Fig. 2-30a as a Cartesian vector.

#### SOLUTION

The angles of  $60^\circ$  and  $45^\circ$  defining the direction of  $\mathbf{F}$  are *not* coordinate direction angles. Two successive applications of the parallelogram law are needed to resolve  $\mathbf{F}$  into its  $x$ ,  $y$ ,  $z$  components. First  $\mathbf{F} = \mathbf{F}' + \mathbf{F}_z$ , then  $\mathbf{F}' = \mathbf{F}_x + \mathbf{F}_y$ , Fig. 2-30b. By trigonometry, the magnitudes of the components are

$$F_z = 100 \sin 60^\circ \text{ N} = 86.60 \text{ N}$$

$$F' = 100 \cos 60^\circ \text{ N} = 50 \text{ N}$$

$$F_x = F' \cos 45^\circ = 50 \cos 45^\circ \text{ N} = 35.36 \text{ N}$$

$$F_y = F' \sin 45^\circ = 50 \sin 45^\circ \text{ N} = 35.36 \text{ N}$$

Realizing that  $\mathbf{F}_y$  is in the  $-\mathbf{j}$  direction, we have

$$\mathbf{F} = \{35.36\mathbf{i} - 35.36\mathbf{j} + 86.60\mathbf{k}\} \text{ N} \quad \text{Ans.}$$

To show that the magnitude of this vector is indeed 100 N, apply Eq. 2-4,

$$\begin{aligned} F &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{35.36^2 + 35.36^2 + 86.60^2} = 100 \text{ N} \end{aligned}$$

If needed, the coordinate direction angles of  $\mathbf{F}$  can be determined from the components of the unit vector acting in the direction of  $\mathbf{F}$ .

$$\begin{aligned} \mathbf{u} &= \frac{\mathbf{F}}{F} = \frac{F_x}{F}\mathbf{i} + \frac{F_y}{F}\mathbf{j} + \frac{F_z}{F}\mathbf{k} \\ &= \frac{35.36}{100}\mathbf{i} - \frac{35.36}{100}\mathbf{j} + \frac{86.60}{100}\mathbf{k} \\ &= 0.3536\mathbf{i} - 0.3536\mathbf{j} + 0.8660\mathbf{k} \end{aligned}$$

so that

$$\alpha = \cos^{-1}(0.3536) = 69.3^\circ$$

$$\beta = \cos^{-1}(-0.3536) = 111^\circ$$

$$\gamma = \cos^{-1}(0.8660) = 30.0^\circ$$

These results are shown in Fig. 2-30c.

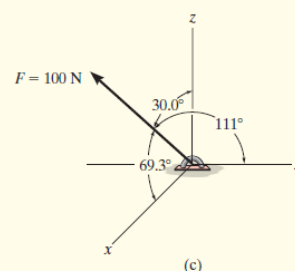
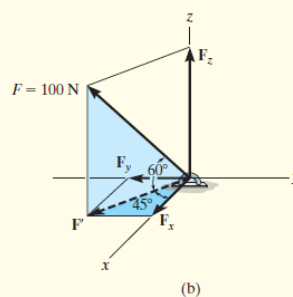
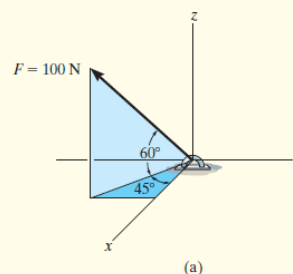


Fig. 2-30

## 1.10 Practice In Classroom

### EXAMPLE 2.14

The frame shown in Fig. 2-42a is subjected to a horizontal force  $\mathbf{F} = \{490\mathbf{j}\}$  N. Determine the magnitudes of the components of this force parallel and perpendicular to member  $AB$ .

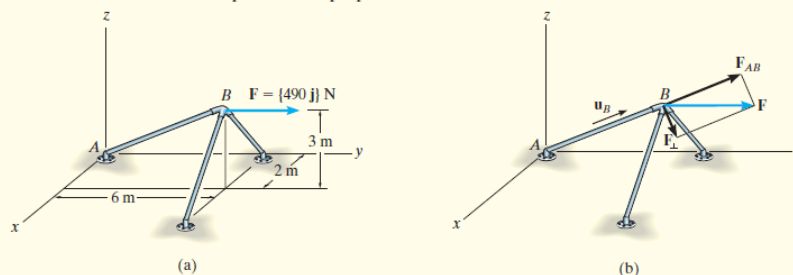


Fig. 2-42

#### SOLUTION

The magnitude of the projected component of  $\mathbf{F}$  along  $AB$  is equal to the dot product of  $\mathbf{F}$  and the unit vector  $\mathbf{u}_{AB}$ , which defines the direction of  $AB$ , Fig. 2-42b. Since

$$\mathbf{u}_{AB} = \frac{\mathbf{r}_{AB}}{r_{AB}} = \frac{2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}}{\sqrt{2^2 + 6^2 + 3^2}} = \frac{2}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{3}{7}\mathbf{k}$$

then

$$F_{AB} = F \cos \theta = \mathbf{F} \cdot \mathbf{u}_{AB} = (490\mathbf{j}) \cdot \left(\frac{2}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{3}{7}\mathbf{k}\right)$$

$$= (0)\left(\frac{2}{7}\right) + (490)\left(\frac{6}{7}\right) + (0)\left(\frac{3}{7}\right)$$

$$= 420 \text{ N} \quad \text{Ans.}$$

Since the result is a positive scalar,  $\mathbf{F}_{AB}$  has the same sense of direction as  $\mathbf{u}_{AB}$ , Fig. 2-42b.

Expressing  $\mathbf{F}_{AB}$  in Cartesian vector form, we have

$$\mathbf{F}_{AB} = F_{AB}\mathbf{u}_{AB} = (420 \text{ N})\left(\frac{2}{7}\mathbf{i} + \frac{6}{7}\mathbf{j} + \frac{3}{7}\mathbf{k}\right)$$

$$= \{120\mathbf{i} + 360\mathbf{j} + 180\mathbf{k}\} \text{ N} \quad \text{Ans.}$$

The perpendicular component, Fig. 2-43b, is therefore

$$\mathbf{F}_{\perp} = \mathbf{F} - \mathbf{F}_{AB} = 490\mathbf{j} - (120\mathbf{i} + 360\mathbf{j} + 180\mathbf{k})$$

$$= \{-120\mathbf{i} + 130\mathbf{j} - 180\mathbf{k}\} \text{ N}$$

Its magnitude can be determined either from this vector or by using the Pythagorean theorem, Fig. 2-42b:

$$F_{\perp} = \sqrt{F^2 - F_{AB}^2} = \sqrt{(490 \text{ N})^2 - (420 \text{ N})^2}$$

$$= 252 \text{ N} \quad \text{Ans.}$$

## 1.11 Further Reading

These topics are covered in Chapter 2 in the Statics & Mechanics of Materials textbook (5<sup>th</sup> Edition, 2018). Make sure you read and understand the following sections:

- Section 2.1 Scalars and Vectors
- Section 2.2 Vector Operations
- Section 2.3 Vector Addition of Forces

- Section 2.4 Addition of A system of Coplanar Forces
- Section 2.5 Cartesian vectors
- Section 2.6 Addition of Cartesian vectors
- Section 2.7 Position vectors
- Section 2.9 Dot product

### **1.12 Work To Do**

Statics & Mechanics of Materials textbook:

Go through the 'Preliminary Problems' and 'Fundamental Problems' of the following sections:

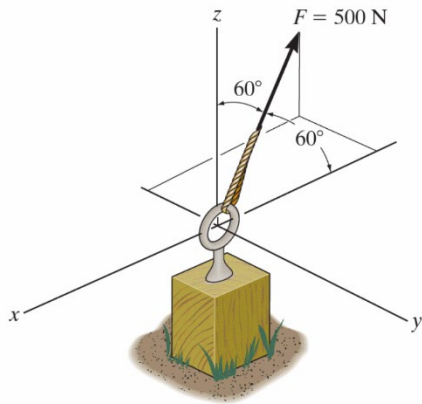
- Section 2.4 Addition of A system of Coplanar Forces
- Section 2.6 Addition of Cartesian vectors
- Section 2.9 Dot product

Solutions and answers to these questions are provided at the end of the textbook.

## Tutorial Questions

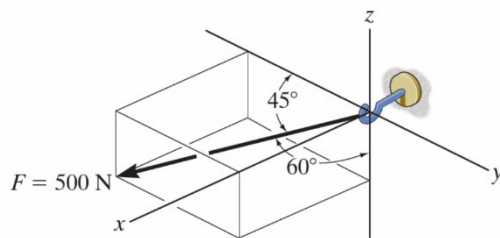
### Question 1

**F2-14.** Express the force as a Cartesian vector.



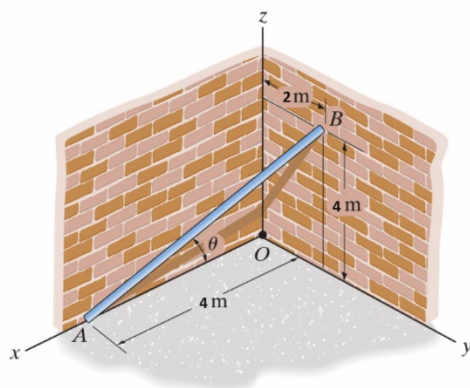
### Question 2

**F2-15.** Express the force as a Cartesian vector.



### Question 3

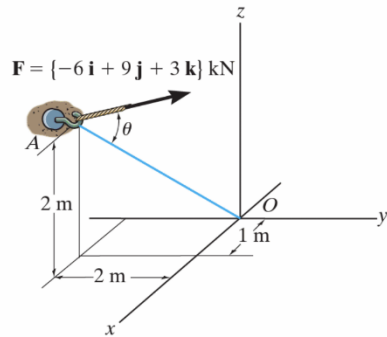
**F2-20.** Determine the length of the rod and the position vector directed from  $A$  to  $B$ . What is the angle  $\theta$ ?





#### Question 4

**F2-25.** Determine the angle  $\theta$  between the force and the line  $AO$ .



**Prob. F2-25**

#### Question 5

**F2-26.** Determine the angle  $\theta$  between the force and the line  $AB$ .

